

Riemann Hypothesis Research Program: Complete Progress Report and Formal Ledger

Logan Eberwein

Independent Researcher

loganeberwein@icloud.com ORCID: [0009-0004-5650-9313](https://orcid.org/0009-0004-5650-9313)

September 2026; revised October 2026

Lean 4 sources: <https://github.com/loganeberwein6/Logan-Eberwein-RH-Research>, branch
stage31-two-channel

Abstract

This document is a complete research ledger for a one-year Riemann Hypothesis investigation, begun in the author’s freshman year of high school, combining Lean 4 formal verification, computational search, and analytic mathematics. All 16 Lean modules of the repository are surveyed; all 190 research markdown documents are synthesized. Every result is labeled with its verification status: **[LEAN-VERIFIED]** (proved in a Lean module declared in the repository build file, no `sorry`), **[LEAN-ARCHIVE]** (formalized in a development stage whose Lean source is not in the public repository), **[PROVED]** (mathematically proved, not yet formalized), **[NUMERICAL]** (verified by computation), **[HEURISTIC]** (plausible derivation, not rigorous), **[KILLED]** (explicitly disproved with counterexample), **[CONJECTURE]** (stated precisely, status open), or **[OPEN]** (load-bearing open problem). The Riemann Hypothesis is **[OPEN]**.

Contents

1	Scope and disclaimer	3
2	Lean-formalized theorems: SR/Rees matrix program (Stages 0–40)	3
2.1	Fundamental definitions	3
2.2	Rank theorem	3
2.3	Threshold matrix: coercivity theorems	4
2.4	Global positivity	4
2.5	Indefiniteness: impossibility theorems	5
2.6	Dagger involution (bidegree analysis)	5
2.7	Coordinate norm positivity (formerly “GNS-type positivity”)	6
2.8	Zero-transfer (margin argument)	6
2.9	Arithmetic bridge and channel decompositions	6
2.10	Dirichlet series identities	7
3	Proved results: analytic program	7
3.1	Weil form contraction search (25+ candidates)	7
3.2	Killed approaches: exact failure theorems	8
3.3	Structural obstruction theorems	9

4	The required operator: formal criteria	9
5	Conjectures: precisely stated open problems	10
6	The frontier	11
7	Repository and reproducibility	11
8	Summary table	11

1 Scope and disclaimer

This is a research report, not a proof of RH. Numerical positivity at finitely many cutoffs does not imply positivity of the limiting Weil form. Heuristic asymptotics do not establish analytic theorems. Every claim is labeled by its exact verification status. The program has identified the precise mathematical frontier; crossing it requires genuinely new analytic content.

2 Lean-formalized theorems: SR/Rees matrix program (Stages 0–40)

The public repository (branch `stage31-two-channel`) contains 16 Lean modules, all declared in its build file (`lakefile.lean`; the build was last recorded as verified in commit `444a9be`). No Lean file in the repository contains `sorry`, `admit`, or an added axiom. Statements proved in these modules are labeled **[LEAN-VERIFIED]**. Statements that were formalized during development but whose Lean source is not in the public repository are labeled **[LEAN-ARCHIVE]**; the September version of this report labeled them **[LEAN-VERIFIED]** (see the revision note at the end).

2.1 Fundamental definitions

Definition 2.1 (Rees comparison matrix). **[LEAN-VERIFIED]** For $X \geq 6$, define $M_{\text{Rees},X} : \text{Fin}(X-2) \times \text{Fin}(X-2) \rightarrow \mathbb{Q}$ by

$$M_{\text{Rees},X}(i, j) = \begin{cases} -1 & (i+2)(j+2) < X \\ +1 & (i+2)(j+2) \geq X. \end{cases}$$

Definition 2.2 (Quotient count and threshold set). **[LEAN-VERIFIED]**

$$\text{SR_quotient_count}(X) = \#\{\lfloor (X-1)/m \rfloor : m \in \{2, \dots, X-1\}\}.$$

The threshold set $\text{SR_threshold_set}(X)$ is the image of $m \mapsto \lfloor (X-1)/m \rfloor$ over $\{2, \dots, X-1\}$.

Definition 2.3 (Threshold submatrix). **[LEAN-VERIFIED]** For strictly monotone $t_0 < t_1 < \dots < t_n$:

$$(S_t)_{ij} = \begin{cases} -1 & t_j \leq t_i \\ +1 & t_j > t_i. \end{cases}$$

Definition 2.4 (von Mangoldt function). **[LEAN-VERIFIED]** $\Lambda(n) = \log p$ if $n = p^k$ for some prime p and $k \geq 1$; else $\Lambda(n) = 0$.

2.2 Rank theorem

Theorem 2.5 (Rees rank formula). **[LEAN-VERIFIED]** (`SR_rees_rank_eq_quotient_count`) For all $X \geq 6$:

$$\text{rank}(M_{\text{Rees},X}) = \text{SR_quotient_count}(X) \approx 2\sqrt{X}.$$

Lemma 2.6 (Row equivalence). **[LEAN-VERIFIED]** (`SR_rees_matrix_row_eq_iff_quotient_eq`) Rows m and m' of $M_{\text{Rees},X}$ are equal iff $\lfloor (X-1)/m \rfloor = \lfloor (X-1)/m' \rfloor$.

Lemma 2.7 (Threshold row difference). *[LEAN-VERIFIED] (SR_rees_threshold_row_diff) For quotients $q_1 < q_2$ and index j :*

$$R_{q_2}(j) - R_{q_1}(j) = \begin{cases} -2 & q_1 < j + 2 \leq q_2 \\ 0 & \text{otherwise.} \end{cases}$$

Theorem 2.8 (Quotient rows are linearly independent). *[LEAN-VERIFIED] (SR_rees_quotient_rows_linearIndependent) The threshold representative rows of $M_{\text{Rees},X}$ are linearly independent over $\mathbb{Q}$.*

2.3 Threshold matrix: coercivity theorems

Theorem 2.9 (Symmetric part identity). *[LEAN-VERIFIED] (SR_threshold_matrix_sym_add_self) For any strictly monotone $t : \text{Fin}(n+1) \rightarrow \mathbb{N}$:*

$$S_t + S_t^T = -2I.$$

Theorem 2.10 (Exact quadratic form). *[LEAN-VERIFIED] (SR_threshold_matrix_quadratic_form)*

$$x^T S_t x = -\|x\|^2 \quad \text{for all } x \in \mathbb{Q}^{n+1}.$$

Theorem 2.11 (Trivial kernel / injectivity). *[LEAN-VERIFIED] (SR_threshold_matrix_mulVec_injective) S_t is injective: $S_t x = 0 \Rightarrow x = 0$.*

Theorem 2.12 (Skew decomposition). *[LEAN-VERIFIED] (SR_threshold_matrix_eq_neg_one_add_skew) $S_t = -I + K$ where $K = (S_t - S_t^T)/2$ is skew-symmetric ($K^T = -K$).*

Corollary 2.13. *[LEAN-VERIFIED] The skew part satisfies $K^T = -K$ (SR_threshold_matrix_skew_transpose).*

Theorem 2.14 (Signed threshold determinant). *[LEAN-VERIFIED] (SR_signed_threshold_matrix_det)*

$$\det(S_t) = (-1) \cdot (-2)^n \neq 0.$$

Theorem 2.15 (Linear independence from determinant). *[LEAN-VERIFIED] (SR_signed_threshold_matrix_linearIndependent) The rows of S_t are linearly independent over $\mathbb{Q}$.*

2.4 Global positivity

Theorem 2.16 (All-ones positivity). *[LEAN-VERIFIED] (allOnesEntrySum_positive_general) For all $X \geq 6$:*

$$A_X := \sum_{m,n \in \{2, \dots, X-1\}} M_{\text{Rees},X}(m,n) > 0.$$

Observation 2.17. [NUMERICAL] Machine-checked: $A_X > 0$ for all $6 \leq X \leq 1000$. Benchmark values: $A_6 = 5$, $A_{10} = 35$, $A_{30} = 635$, $A_{100} = 8855$, $A_{210} = 41381$, $A_{2310} = 5295017$.

Theorem 2.18 (Hyperbola reduction). *[LEAN-ARCHIVE] $A_X = (X-2)^2 - 2N_{\text{int}}(X)$ where $N_{\text{int}}(X) = \#\{(m,n) : mn < X, m,n \in \text{support}\}$. The positivity follows from $N_{\text{int}}(X) < (X-2)^2/2$ via the Dirichlet hyperbola method.*

2.5 Indefiniteness: impossibility theorems

These are among the most important results: they rule out entire classes of proof strategies.

Theorem 2.19 (Negative direction exists). *[LEAN-VERIFIED]* (SR_form_has_negative_direction) *For all $X \geq 6$, there exists $v : \text{Fin}(X - 2) \rightarrow \mathbb{R}$ such that*

$$\text{SR_quadratic_form}(X, v) < 0.$$

The basis vector e_0 (supported at index 0, which corresponds to $m = 2$) witnesses this.

Theorem 2.20 (Raw SR form is not PSD). *[LEAN-VERIFIED]* (SR_form_not_positive_semidefinite) *For all $X \geq 6$:*

$$\neg (\forall v, 0 \leq \text{SR_quadratic_form}(X, v)).$$

Observation 2.21 (No PSD realization preserving raw form). *[LEAN-VERIFIED]* (SR_no_psd_realization_preserving_raw_form) Let $X \geq 6$. There is no positive semidefinite form P satisfying $P(v) = \text{SR_quadratic_form}(X, v)$ for all v . Formally: for any $P : (\text{Fin}(X - 2) \rightarrow \mathbb{R}) \rightarrow \mathbb{R}$ with $P(v) \geq 0$ everywhere, $P \neq \text{SR_quadratic_form}(X, \cdot)$. This statement is true but tautological: no everywhere-nonnegative function can equal a function that takes negative values. It does not exclude compressions, restrictions, or other transformations of the form, and is therefore recorded as an observation rather than a theorem.

Theorem 2.22 (Global coercivity is false). *[LEAN-VERIFIED]* (SR_global_coercivity_false)

$$\neg (\forall X \geq 6, \forall v, 0 \leq \text{SR_quadratic_form}(X, v)).$$

Theorem 2.23 (SR form is indefinite). *[LEAN-VERIFIED]* (SR_form_is_indefinite_of_positive) *For all $X \geq 6$: whenever the SR quadratic form takes a positive value, it also takes a negative value. The form is indefinite.*

Theorem 2.24 (Explicit $X = 6$ witness). *[LEAN-VERIFIED]* (SR_X6_indefinite) *At $X = 6$: $\text{SR_quadratic_form}(6, e_0) < 0$ and $\text{SR_quadratic_form}(6, \mathbf{1}) = 14 > 0$.*

Theorem 2.25 (Uniform subspace is positive). *[LEAN-VERIFIED]* (SR_uniform_X6_positive, SR_uniform_X6_strict) *For $X = 6$, on the 1-dimensional subspace of constant vectors: $\text{SR_quadratic_form}(6, c \cdot \mathbf{1}) = 14c^2 \geq 0$ with equality iff $c = 0$.*

2.6 Dagger involution (bidegree analysis)

Theorem 2.26 (Dagger is an involution). *[LEAN-VERIFIED]* (dagger_involution) *On the $(1, 1)$ carrier E^{11} at $X = 6$: $\dagger(\dagger(x)) = x$ for all x .*

Theorem 2.27 (Dagger swaps labels). *[LEAN-VERIFIED]* (dagger_h_swap) *For all m, n : $\dagger(h_{m\bar{n}}) = h_{n\bar{m}}$.*

Theorem 2.28 (Diagonal elements are fixed). *[LEAN-VERIFIED]* (dagger_diagonal_fixed) *For all m : $\dagger(h_{m\bar{m}}) = h_{m\bar{m}}$.*

Theorem 2.29 (Dagger is $\mathbb{C}$ -antilinear). *[LEAN-VERIFIED]* $\dagger(x+y) = \dagger(x) + \dagger(y)$ and $\dagger(a \cdot x) = \bar{a} \cdot \dagger(x)$.

2.7 Coordinate norm positivity (formerly “GNS-type positivity”)

Remark 2.30. The Lean names retain the prefix `GNS`, but these statements reduce to $|x_{22}|^2 \geq 0$; they are coordinate norm positivity lemmas, not a GNS reconstruction.

Theorem 2.31 (Coordinate norm positivity). *[LEAN-VERIFIED] (GNS_positive) For all $x \in E^{11}$: $0 \leq |\Gamma_{\text{obs}}(x)|^2$.*

Theorem 2.32 (Dagger coordinate positivity). *[LEAN-VERIFIED] (GNS_positive_correct) For all $x \in E^{11}$: $0 \leq \text{Re}(\Gamma_{\text{obs}}(\dagger(x) \cdot x))$.*

Theorem 2.33 (Type-level firewall). *[LEAN-VERIFIED] (Gamma_obs_domain_E11) The coordinate functional Γ_{obs} is defined on E^{11} , not on E_6^{20} . This is a type-level obstruction: the (1,1)-sector extension principle is not automatic.*

2.8 Zero-transfer (margin argument)

Theorem 2.34 (Nonzero transfer). *[LEAN-VERIFIED] (nonzero_transfer_of_margin) In any normed ring: if $\delta > 0$, $\delta \leq \|F\|$, and $\|G - F\| < \delta$, then $G \neq 0$.*

Theorem 2.35 (Uniform zero-free transfer). *[LEAN-VERIFIED] (uniform_zero_free_transfer) Pointwise: if $\delta \leq \|F(x)\|$ and $\|G(x) - F(x)\| < \delta$ for all x , then $G(x) \neq 0$ for all x .*

2.9 Arithmetic bridge and channel decompositions

Definition 2.36 (SR Sint and Sext). *[LEAN-VERIFIED]*

$$S_{\text{int},\log}(X) = \sum_{\substack{m,n \in \text{supp}(X) \\ mn < X}} \frac{\log m \cdot \log n}{\sqrt{mn}},$$

$$S_{\text{ext},\log}(X) = \sum_{\substack{m,n \in \text{supp}(X) \\ mn \geq X}} \frac{\log m \cdot \log n}{\sqrt{mn}}.$$

Theorem 2.37 (B3 bound). *[LEAN-VERIFIED] (B3_final, Stage 27)*

$$\frac{S_{\text{int},\log}(X)}{T_{\log}(X)} \leq \frac{1 - \sqrt{\kappa_{\log}(X)}}{2}.$$

Theorem 2.38 (Kappa bounds). *[LEAN-VERIFIED] $\kappa_{\log}(X) \in [0, 1]$ for all X .*

Theorem 2.39 (Two-channel decomposition). *[LEAN-VERIFIED] (SR_two_channel_decomposition, Stage 28)*

$$Q = \frac{1}{2}Q_{\text{ratio}} + \frac{1}{2}Q_{\text{product}}$$

via the identity $\cos A \cos B = \frac{1}{2} \cos(A - B) + \frac{1}{2} \cos(A + B)$.

Theorem 2.40 (Ratio channel diagonal bound). *[LEAN-ARCHIVE] (SR_ratio_diagonal_bound, Stage 33) The diagonal of Q_{ratio} satisfies:*

$$\text{diag}(Q_{\text{ratio}})/X^{2\beta} \rightarrow \frac{1}{2\beta} \quad \text{as } X \rightarrow \infty.$$

2.10 Dirichlet series identities

Theorem 2.41 (von Mangoldt L-series). *[LEAN-ARCHIVE]* (sr37_clean_vonMangoldt_lseries_eq_negative_zeta_log_deriv) *In the half-plane $\text{Re}(s) > 1$:*

$$\sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s} = -\frac{\zeta'(s)}{\zeta(s)}.$$

This identity is a one-line application of Mathlib's `ArithmeticFunction.LSeries_vonMangoldt_eq_deriv_riemannZeta_div`; it is not an original contribution of this program.

Theorem 2.42 (Convolution L-series). *[LEAN-ARCHIVE]* (sr37_clean_unrestricted_pair_convolution_lseries_eq_zeta_sq)

$$L(\mathbf{1} * \mathbf{1}, s) = \zeta(s)^2.$$

Theorem 2.43 (Finite Mellin kernels). *[LEAN-ARCHIVE]* (Stage 36) *Sharp-cutoff finite Mellin transform $\rightarrow 1/s$. First-difference finite Mellin transform $\rightarrow 1/(s(s+1))$. Second-difference finite Mellin transform $\rightarrow 2/(s(s+1)(s+2))$.*

Theorem 2.44 (SR Dirichlet series limit). *[PROVED]* *As $X \rightarrow \infty$, the SR Dirichlet series converges to:*

$$D_{\text{SR}}(s) \rightarrow -(\zeta(s) - 1)^2 \quad (\text{support starts at } m = 2).$$

With full support $m \geq 1$: $D_{\text{full}}(s) \rightarrow -\zeta(s)^2$.

Theorem 2.45 (Signature data at primorials). *[NUMERICAL]* *The Rees matrix signature (p, q, r) is balanced ($p = q$) at primorial values $X = 6, 30, 210, 2310$ under complete integer support. Verified: $X = 6, 30, 210, 2310$ all give $p = q$. Non-primorials $X = 7, 35$ give $p \neq q$.*

3 Proved results: analytic program

3.1 Weil form contraction search (25+ candidates)

Theorem 3.1 (Hardy-Gram exact multiplicativity). *[PROVED]* *For all positive integers c, m, n :*

$$G_H(cm, cn) = c \cdot G_H(m, n),$$

where

$$G_H(m, n) = \frac{\sqrt{mn}}{2\pi} \int_{-\infty}^{\infty} \left(\frac{m}{n}\right)^{it} \frac{|\zeta(\frac{1}{2} + it)|^2}{\frac{1}{4} + t^2} dt.$$

Proof: $\sqrt{cm} \cdot \sqrt{cn} = c\sqrt{mn}$ and $(cm/cn)^{it} = (m/n)^{it}$. $\square$

Corollary 3.2. *[PROVED]* $G_H(m, n) = \text{gcd}(m, n) \cdot G_H(m/\text{gcd}, n/\text{gcd})$.

Theorem 3.3 (Hardy-Gram structural form). *[PROVED]*

$$G_H(m, n) = \sqrt{mn} \cdot H\left(\log \frac{m}{n}\right), \quad H(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{|\zeta(\frac{1}{2} + it)|^2}{\frac{1}{4} + t^2} e^{i\tau t} dt.$$

H is the Fourier transform of $|\zeta(1/2 + it)|^2/(1/4 + t^2)$.

Theorem 3.4 (Hardy-Gram asymptotics). *[HEURISTIC] From the approximate functional equation: as the integration cutoff $T \rightarrow \infty$,*

$$\frac{G_H(m, n)}{G_H(1, 1)} \rightarrow \gcd(m, n).$$

The normalized Hardy Gram satisfies $G_H(m, n)/\sqrt{G_H(m, m)G_H(n, n)} \rightarrow \gcd(m, n)/\sqrt{mn}$.

Observation 3.5. *[NUMERICAL] Diagonal: $G_H(n, n)/G_H(1, 1) = n$ exactly (by multiplicativity). Off-diagonal: $G_H(1, 2)/G_H(1, 1)$ decreases from 1.230 at $T = 100$ to 1.204 at $T = 1000$; these values do not establish convergence to $\gcd(1, 2) = 1$.*

Theorem 3.6 (GCD kernel failure margin). *[NUMERICAL] The Weil form contraction norm with the GCD kernel satisfies:*

$$\|C_L\| - 1 \sim L^{7.30} \quad (L \in [6.5, 8.0], \text{ 4-point power-law fit}).$$

Theorem 3.7 (Dominant failure modes at small primes). *[NUMERICAL] The dominant negative eigenvector of $G_L^*G_L - F_L^*F_L$ is concentrated on prime-power channels with $n = 2, 4, 7, 9, 13, 16, 17, 19$ consistently across $L = 6, 7, 8$. These are NOT boundary primes near $\exp(2L)$.*

3.2 Killed approaches: exact failure theorems

Theorem 3.8 (SR coefficients $\neq \Lambda$). *[PROVED] (counterexample) $\text{SR_coeff}(6, 4) = -1 \neq \Lambda(4) = \log 2$. SR produces divisor-count coefficients, not von Mangoldt weights.*

Theorem 3.9 (Abel bridge wrong scale). *[PROVED] (Stage 39 disproof) $S_{\text{int}, \log}(X) \sim 2\sqrt{X} \log X$, while $2\psi_2(X)/\sqrt{X} \sim X^{3/2}$. These differ by a factor of X . The Abel bridge approximation is false.*

Theorem 3.10 (ζ^2 gives zeros not poles at ζ zeros). *[PROVED] $D_{\text{full}}(s) = -\zeta(s)^2$ has zeros of order 2 at ζ zeros, not poles. Perron/explicit formula gives no contribution from ζ zeros.*

Theorem 3.11 (Weil/PW clamped inequality is false for small L). *[PROVED] For $h(y) = y(1 - y^2)^2$: as $L \rightarrow 0$, $R_L \rightarrow 337/105 < \infty$ while $c_L \sim \pi^2/(2L) \rightarrow \infty$. For all $0 < L \leq 1/10$: $R_L < 4$ while $c_L > 20$. The inequality $E_L[HF] \geq c_L \|HF\|^2$ is false for all sufficiently small L .*

Theorem 3.12 (Relaxed-source scaling obstruction). *[PROVED] (Research Ledger) For $D = ch$ and any positive scaling $\lambda > 0$: the LP inequality $\lambda^2(e - \ell^2) + 8\lambda\ell - 16 \leq 0$ fails for large positive λ whenever $e > \ell^2$. Any theorem extending contractivity to arbitrary scaled sources is false.*

Theorem 3.13 (Volterra endpoint mismatch). *[PROVED] (Research Ledger) The causal convolution representation V_α satisfies $V_{\delta_T} = 0$ as an operator on $L^2(0, T)$. Hence $V_{\nu_M} = V_{\mu_M}$ despite cos-channel difference $V_M(0) = C_M(0) - m_M \cos(0) \neq 0$. The Volterra representation is non-faithful on terminal boundary mass. Smallest counterexample: $\alpha = \delta_T$ with $0 < M \leq (\log 2)/2$.*

Theorem 3.14 (Finite moment closure killed). *[PROVED] (Research Ledger) A polynomial identity in the endpoint-centered moments $b_k = \int (T - s)^k d\mu(s)$ that closes the terminal ladder at any finite order would require a polynomial relation vanishing on the continuum support interval — hence the zero polynomial. Therefore the exact linear scale/endpoint state cannot be finite-dimensional.*

Theorem 3.15 (Ratio-channel GUE detection is γ -independent). *[PROVED]* The matrix $M_{\text{Rees}}(m, n) \cdot e^{i\gamma(\log m - \log n)}$ is unitarily equivalent (via diagonal conjugation) to M_{Rees} for every γ . Its spectrum, and hence its r -statistic, is γ -independent.

Theorem 3.16 (Product channel non-specificity). *[NUMERICAL]* The product-channel kernel $\text{Rees}(m, n) \cdot \cos(\gamma \log(mn)/2)$: at $X = 200$, $\gamma = 35.0$ (not a ζ zero) gives $r = 0.6054$, closer to GUE ($= 0.5996$) than any tested ζ zero frequency.

Theorem 3.17 (Möbius-log diagonal realization is indefinite). *[PROVED]* The specific diagonal realization with entries $\mu(m) \log m$ is indefinite. Note $\mu(2) = -1$ and $\mu(30) = -1$, so both $\mu(2) \log 2 < 0$ and $\mu(30) \log 30 < 0$. The claim is narrow: this particular realization is indefinite. It does not follow that any operator with trace $\sum_k \Lambda(k)/k^s$ must be indefinite, since trace alone does not determine operator signature.

Theorem 3.18 (BN target vector is tautological in gcd model). *[PROVED]* In the gcd kernel model with $b_k = 1/\sqrt{k}$: b = first column of G , so $G^{-1}b = e_1$ and $b^T G^{-1}b = G_{11} = 1$. Hence $d_N = 0$ for all N by linear algebra alone. This is not evidence of BN convergence.

Theorem 3.19 (Arakelov matrix is indefinite). *[NUMERICAL]* The matrix $\text{Ar}(m, n) = \frac{1}{2} \log(\text{lcm}(m, n) / \gcd(m, n)^2) \cdot \gcd(m, n) / \sqrt{mn}$ has minimum eigenvalue ≈ -5.49 at $N = 100$ with 99 negative eigenvalues. It is strongly indefinite.

3.3 Structural obstruction theorems

Theorem 3.20 (Type mismatch). *[PROVED]* RH is a support condition: zeros lie on $\text{Re}(s) = 1/2$. Matrix positivity is a sign condition: eigenvalues ≥ 0 . These require fundamentally different mathematical machinery: support conditions need global analytic continuation; sign conditions are local and finite. The claim is narrow: this specific SR comparison matrix has no currently proved mechanism converting its finite sign information into the required analytic support statement. Note that Li's criterion is a positivity condition equivalent to RH , and the Redheffer matrix determinant equals the Mertens function—so finite matrices can encode RH -equivalent information.

Theorem 3.21 (Separability obstruction). *[PROVED]* $\Lambda(mn) \neq \Lambda(m)\Lambda(n)$ for general m, n . No separable kernel $f(m)g(n)$ applied to any matrix with entries depending separately on m and n can produce Λ as a coefficient sequence.

Theorem 3.22 (Circularity obstruction). *[PROVED]* The Beurling-Nyman target $b_n = \langle 1_{[0,1]}, \rho_n \rangle_{H^2}$ requires evaluating $\zeta(1/2 + it)$ on the critical line. In our specific construction this requires knowing the zeros of ζ , creating a circular dependency. This does not mean $\zeta(1/2 + it)$ cannot be evaluated unconditionally — it can — but that our particular BN target representation depends on the zero locations in a way we have not resolved.

Theorem 3.23 (Arithmetic/spectral duality obstruction). *[PROVED]* (Research Ledger) The Weil explicit formula's arithmetic side (primes, poles, gamma) and spectral side (zeros) are dual descriptions of the same functional. They are not independent positive-energy terms that may be added. Any attempt to supplement the arithmetic form with zero-contribution energy changes the form rather than improving the original one.

4 The required operator: formal criteria

Definition 4.1 (Six required properties for the correct operator). An operator $T(s)$ could prove RH if it satisfies all of:

P1 [Poles at zeros] $\det(I - T(s)) = 0$ at each ζ zero ρ .

P2 [Prime-power weights] $\text{Tr}(T^n(s))$ counts prime powers with weight $\Lambda(p^k)$.

P3 [Positivity] Some canonical quadratic form $Q[T] \geq 0$.

P4 [Functional equation] $T(s)$ and $T(1-s)$ are related; $\text{Re}(s) = 1/2$ is the fixed line.

P5 [Arithmetic origin] T is constructible from integers; finite truncations $T_X \rightarrow T$.

P6 [Convergence] $\det(I - T(s))$ is well-defined and analytically continuable through the critical strip.

Theorem 4.2 (Property table). [NUMERICAL]/[PROVED]

Operator	P1	P2	P3	P4	P5	P6
SR Rees $M_{\text{Rees}, X}$	×	×	~	×	✓	×
Full-support SR	×	×	~	×	✓	×
Threshold submatrix S_t	×	×	✓	×	✓	×
Weil/PW $H = -D^2 + 1/4$	~	×	~	✓	×	✓
Mayer transfer $\mathcal{L}_s$	✓	✓	×	✓	×	✓
Connes adelic $L^2(A_{\mathbb{Q}}/\mathbb{Q}^{\times})$	✓	✓	?	✓	×	✓
GCD matrix $\text{gcd}(m, n)/\sqrt{mn}$	~	✓	✓	×	✓	×
Hardy-Gram G_H (normalized)	✓	~	✓	~	~	~

No existing operator satisfies all six properties.

5 Conjectures: precisely stated open problems

Conjecture 5.1 (Primorial balance). [CONJECTURE] The Rees matrix signature satisfies $p = q$ (balanced) at every primorial $X = P_k = 2 \cdot 3 \cdot 5 \cdots p_k$ under complete integer support $\{1, \dots, X-1\}$. Verified: $k = 2, 3, 4$ ($X = 6, 30, 210, 2310$). Open for all k .

Conjecture 5.2 (Universal signature bound). [CONJECTURE] For all $X \geq 6$: $|\text{pos}(X) - \text{neg}(X)| \leq 1$, where pos and neg count positive and negative eigenvalues of $M_{\text{Rees}, X}$. Verified numerically for $X \leq 1000$.

Conjecture 5.3 (Dagger-completion defect). [CONJECTURE] The fundamental obstruction across all major RH reformulations is the dagger-completion defect: the inability to convert forward multiplicative structure (A^2) into Hermitian/dagger structure ($A^\dagger A$). This gap, denoted $\mathfrak{D}_+(A)$, is conjectured to be the universal failure invariant. Status: identified across SR, Connes, Weil programs. Not yet formalized.

Conjecture 5.4 (Hardy-Gram GCD asymptotics). [CONJECTURE]

$$\frac{G_H(m, n)}{G_H(1, 1)} \rightarrow \text{gcd}(m, n) \quad \text{as } T \rightarrow \infty,$$

rigorously, from the asymptotic theory of the twisted second moment of ζ . The heuristic derivation from the approximate functional equation is given in §3. A rigorous proof would follow from the Montgomery-Vaughan mean value theorem for the twisted second moment.

6 The frontier

The following is the load-bearing open equation from the research ledger.

Theorem 6.1 (Load-bearing open equation). *[OPEN] (Research Ledger) Construct from the completed arithmetic source a contractive system whose first derivative maps a structurally derived input b to*

$$x_{\text{obs}} = (\sqrt{B} D, a \mapsto \sqrt{B} P(\Theta_a V_M))$$

and whose independently derived input norm is exactly $\|b\| = |4 - L|$. A generic Schur factorization with these properties is merely equivalent to LP and is not evidence for it.

The three-step frontier:

1. **Step 1** (Requires audit first): Certify or kill the Hardy-Gram asymptotics (Conjecture 4.4). The conjecture $G_H(m, n)/G_H(1, 1) \rightarrow \gcd(m, n)$ is not yet supported by convergent numerics: the ratio $G_H(1, 2)/G_H(1, 1) \approx 1.204$ at $T = 1000$ remains far from the conjectured limit of 1. Resolve before treating this as a connection between the Weil contraction and BN programs.
2. **Step 2** (Open): Compute the correct BN target vector $b_n = \langle 1_{[0,1]}, \rho_n \rangle_{H^2}$ analytically, without assuming RH.
3. **Step 3** (= RH): Prove $d_N \rightarrow 0$, i.e., the Beurling-Nyman approximation converges. This is equivalent to RH by the Báez-Duarte criterion. The gap between Steps 2 and 3 is the Riemann Hypothesis itself.

7 Repository and reproducibility

- **Repository:** <https://github.com/loganeberwein6/Logan-Eberwein-RH-Research>
- **Branch:** stage31-two-channel
- **Lean files:** 16 modules, all declared in `lakefile.lean`; no `sorry`, `admit` or `added axiom` in any file.
- **Documented approaches:** 190 markdown files in `approaches/`
- **Build:** Set `ELAN_HOME` to your `elan` installation path, then `lake update` and `lake build`.
- **Zenodo DOI:** 10.5281/zenodo.22783121

8 Summary table

Result	Status	Stage
Rees matrix definition and basic properties	[LEAN-VERIFIED]	0–10
$\text{rank}(M_{\text{Rees}, X}) = \text{SR_quotient_count}(X)$	[LEAN-VERIFIED]	38
$S_t + S_t^T = -2I$	[LEAN-VERIFIED]	38
$x^T S_t x = -\ x\ ^2$	[LEAN-VERIFIED]	38
S_t injective	[LEAN-VERIFIED]	38

$\det(S_t) = (-1)(-2)^n$	[LEAN-VERIFIED]	38
Threshold rows linearly independent	[LEAN-VERIFIED]	38
$A_X > 0$ for all $X \geq 6$	[LEAN-VERIFIED]	21–23
$A_X > 0$ machine-checked to $X = 1000$	[NUMERICAL]	23
Raw SR form is not PSD	[LEAN-VERIFIED]	32
No PSD realization preserves raw SR form	[LEAN-VERIFIED]	32
Global coercivity is false	[LEAN-VERIFIED]	32
SR form is indefinite	[LEAN-VERIFIED]	32
$X = 6$ explicit indefiniteness witness	[LEAN-VERIFIED]	32
Dagger involution on E^{11}	[LEAN-VERIFIED]	39–40
Dagger coordinate positivity	[LEAN-VERIFIED]	39–40
Type-level domain firewall	[LEAN-VERIFIED]	40
Zero-transfer margin theorem	[LEAN-VERIFIED]	40
$L(\Lambda, s) = -\zeta'/\zeta$	[LEAN-ARCHIVE]	37
$L(\mathbf{1} * \mathbf{1}, s) = \zeta(s)^2$	[LEAN-ARCHIVE]	37
B3 bound: $S_{\text{int}}/T \leq (1 - \sqrt{\kappa})/2$	[LEAN-VERIFIED]	27
$\kappa \in [0, 1]$	[LEAN-VERIFIED]	27
Two-channel decomposition	[LEAN-VERIFIED]	28
Finite Mellin kernels	[LEAN-ARCHIVE]	36
Primorial balance at $X = 6, 30, 210, 2310$	[NUMERICAL]	19–23
Signature bound $ \text{pos} - \text{neg} \leq 1$ ($X \leq 1000$)	[NUMERICAL]	17–23
$G_H(cm, cn) = c \cdot G_H(m, n)$	[PROVED]	This work
$G_H(m, n) = \sqrt{mn} H(\log(m/n))$	[PROVED]	This work
$G_H(m, n)/G_H(1, 1) \rightarrow \gcd(m, n)$	[HEURISTIC]	This work
Weil failure margin $\sim L^{7.3}$	[NUMERICAL]	This work
Small-prime failure mode identification	[NUMERICAL]	This work
Weil/PW clamped inequality false for small L	[PROVED]	This work
Abel bridge wrong scale	[PROVED]	Stage 39
ζ^2 gives zeros not poles	[PROVED]	This work
Ratio-channel GUE γ -independence	[PROVED]	This work
Möbius-log operator indefinite	[PROVED]	This work
BN gcd model is tautological	[PROVED]	This work
Type mismatch obstruction	[PROVED]	This work
Separability obstruction	[PROVED]	This work
Circularity obstruction	[PROVED]	This work
Volterra endpoint mismatch	[PROVED]	Research Ledger
Finite moment closure impossible	[PROVED]	Research Ledger
Relaxed-source scaling failure	[PROVED]	Research Ledger
Arakelov matrix indefinite	[NUMERICAL]	This work
Dagger-completion defect conjecture	[CONJECTURE]	Ledger
Primorial balance conjecture	[CONJECTURE]	Stage 19
Universal signature bound conjecture	[CONJECTURE]	Stage 17
Hardy-Gram GCD asymptotics	[CONJECTURE]	This work
RH	[OPEN]	—

Authorship and AI tools

I set the research directions of this program, including the research-ledger structure, the dagger-completion defect hypothesis and the FICS methodology framework. Lean formalization was carried out with AI tools (ChatGPT, Claude, and Claude Code for build verification), and the computational search used `rh_candidate_search.py` (Python/NumPy). ChatGPT and Claude are listed as tools, not authors.

Revision note (October 2026)

This revision corrects the September 2026 version.

1. The Lean status of every statement now matches the repository. Statements whose Lean source is not in the public repository (the von Mangoldt and convolution L-series identities, the finite Mellin kernels, the ratio-channel diagonal bound and the hyperbola reduction) are now labeled `[LEAN-ARCHIVE]`; the previous version labeled them `[LEAN-VERIFIED]`. The previous version also described the repository as containing 7 Lean files (it contains 16 modules, all declared in the build file) and gave two different commit hashes (`ef2586f` on the title page, `9bbaeb4` in the repository section).
2. The Möbius example is corrected: $\mu(2) = -1$, so $\mu(2) \log 2 < 0$ (the previous version stated the opposite sign).
3. The “no PSD realization” statement is tautological and is now an observation.
4. The “GNS” positivity statements are coordinate norm positivity lemmas and are renamed accordingly (the Lean identifiers are unchanged).
5. The von Mangoldt L-series identity is credited to Mathlib.
6. The Hardy–Gram numerics are no longer described as converging; the type-mismatch and circularity obstructions are stated in their narrow form.
7. The module `SR_Mollifier` contained four placeholder declarations of type `True` that name open conjectures (`SR24_LOG_WEIGHTED_X6_OPEN`, `SR24_HALF_WEIGHTED_X6_OPEN`, `SR24_LOG_WEIGHTED_POSITIVITY_CONJECTURE`, `SR24_HALF_WEIGHTED_POSITIVITY_CONJECTURE`); they proved nothing, were never counted as results, and have been replaced by comments.
8. The acknowledgments are rewritten.
9. The year of Burnol’s note is corrected to 2000.

References

- [1] L. Báez-Duarte, A strengthening of the Nyman–Beurling criterion for the Riemann Hypothesis, *Atti Accad. Naz. Lincei Cl. Sci. Fis. Mat. Natur. Rend. Lincei* **14** (2003), 5–11.
- [2] A. Beurling, A closure problem related to the Riemann zeta-function, *Proc. Nat. Acad. Sci. USA* **41** (1955), 312–314.
- [3] J.-F. Burnol, Sur les formules explicites I: analyse invariante, *C. R. Acad. Sci. Paris Sér. I* **331** (2000), 423–428.

- [4] A. Connes, Trace formula in noncommutative geometry and the zeros of the Riemann zeta function, *Selecta Math. (N.S.)* **5** (1999), 29–106.
- [5] S. Hong and R. Loewy, Asymptotic behavior of eigenvalues of GCD matrices, *Glasg. Math. J.* **46** (2004), 551–569.
- [6] D. Mayer, The thermodynamic formalism approach to Selberg’s zeta function for $\mathrm{PSL}(2, \mathbb{Z})$, *Bull. Amer. Math. Soc.* **25** (1991), 55–60.
- [7] H. L. Montgomery, The pair correlation of zeros of the zeta function, in *Analytic Number Theory*, Proc. Sympos. Pure Math. XXIV, AMS, 1973, 181–193.
- [8] B. Nyman, On the one-dimensional translation group and semi-group in certain function spaces, PhD thesis, Uppsala University, 1950.
- [9] B. Rodgers and T. Tao, The de Bruijn–Newman constant is non-negative, *Forum Math. Pi* **8** (2020), e6.
- [10] A. Weil, Sur les “formules explicites” de la théorie des nombres premiers, *Comm. Sém. Math. Univ. Lund* (tome supplémentaire dédié à Marcel Riesz), 1952, 252–265.
- [11] The Mathlib Community, Mathlib4, <https://github.com/leanprover-community/mathlib4>, 2023.